

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code.	Yes	Proper indentation is maintained throughout the Python, HTML, and CSS files.
2	Proper File Structure	Files and folders are logically organized.	Yes	Project files are organized into folders such as Dataset, Notebook, Model, Templates, Static, and Flask Application.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve readability and maintenance.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are clearly named according to their responsibilities.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments explaining preprocessing, training, and prediction logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, model training, prediction, and web application are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code has been removed and reusable functions are utilized.
8	Error Handling	Exceptions and errors are handled gracefully.	Partial	Basic validation and exception handling are implemented in the Flask application.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Data Preprocessing Module	Python, Pandas	Model Training & Prediction	High
2	Feature Engineering Module	Python	Training Pipeline	High
3	Machine Learning Prediction Module	Scikit-learn, XGBoost	Flask Web Application	High
4	Flask Application	Python, Flask	Real-time Prediction	High
5	HTML Templates	HTML, CSS, Bootstrap	User Interface	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for datasets, models, templates, and application code.
Readability	5	Clear naming conventions, proper formatting, and understandable logic.
Reusability	5	Modular functions allow reuse across preprocessing, training, and prediction stages.
Documentation / Comments	4	Sufficient comments are included to explain major functions and workflow.
Overall Score	4.8 / 5	The codebase is maintainable, readable, modular, and suitable for future enhancements.